

Lecture 2

Shashank Deep

Making a non-spontaneous
process spontaneous or Vice-
Versa

Non-Spontaneous



ΔG

$$\Delta G = \Delta H - T\Delta S$$

Spontaneous

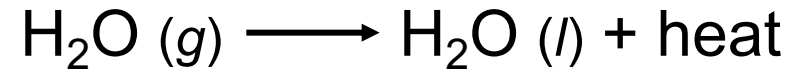
Heat in a reaction or process

A chemical reaction or a physical process is accompanied by absorption or release of heat by the system.

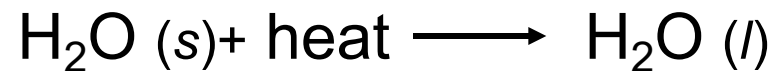
The heat evolved or absorbed in a chemical reaction or a physical process can be measured by measuring heat at constant pressure ($q = \Delta H$).

Exothermic process is a process in which heat is released leading to transfer of thermal energy from the system to the surroundings.

On carrying out an exothermic reaction in a vessel, the reaction vessel will feel warm.



Endothermic process is a process in which heat is absorbed by the system from the surroundings.



Enthalpy of Reaction

The *change* in enthalpy, ΔH , is the enthalpy of the products minus the enthalpy of the reactants:

$$\Delta H = H_{\text{products}} - H_{\text{reactants}}$$

For Endothermic Reaction, $H_{\text{products}} > H_{\text{reactants}}$, $\Delta H > 0$

For Exothermic Reaction, $H_{\text{products}} < H_{\text{reactants}}$, $\Delta H < 0$

A Physical Process



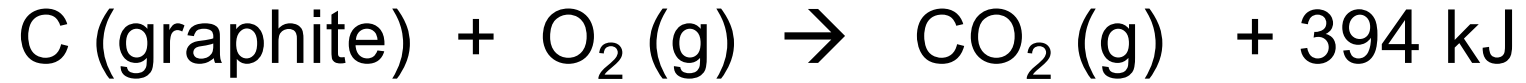
Heat of Vaporization= $\Delta H_{\text{vap}} = 44 \text{ kJ/mole}$

Reaction is endothermic

For the reverse process:

$\Delta H = - 44 \text{ kJ/mole}$, Reaction is exothermic

A Chemical Process



$$\Delta H = -394 \text{ kJ/mole}$$

Reaction is exothermic

Exothermic Chemical Process



Any combustion reaction



Rusting of iron (Used in chemical hot pack)



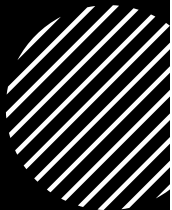
A reaction between water and calcium chloride



A reaction between sodium and chlorine to make sodium chloride



Exothermic Physical process



Dissolving laundry detergent in water



Crystallization of sodium acetate or "hot ice"



Freezing water into ice (fusion)



Snow forming inside clouds



Endothermic Chemical Reaction



Dissolving ammonium chloride in water

Mixing water and ammonium nitrate
(Used in cold pack)

Reaction of barium hydroxide octahydrate
crystals with dry ammonium chloride

Reaction of thionyl chloride (SOCl_2) with
cobalt(II) sulfate heptahydrate

Endothermic Physical processes

Melting of ice cubes

Melting of solid salts

Evaporating liquid water

Conversion frost to water
vapor

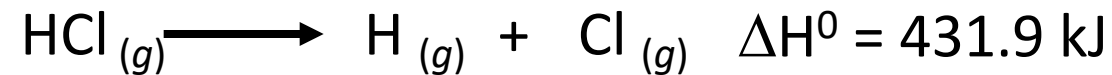
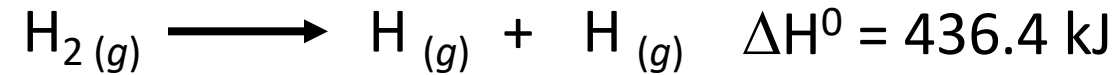
Breaking Bond

- If we want to break a bond between two atoms, we need to apply force to overcome the attractive force.
 - Bond breaking is an endothermic process
-
- Bond may be a covalent bond, electrostatic or van der Waal's bond.

Formation of a bond

- Energy is released when new chemical bonds are formed.
- Bond making is an exothermic process.
- Breaking or making the same chemical bond will require the same energy to be put in or released.

The enthalpy change required to break a particular bond in one mole of gaseous molecule is the ***bond energy***.

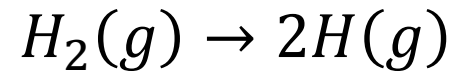
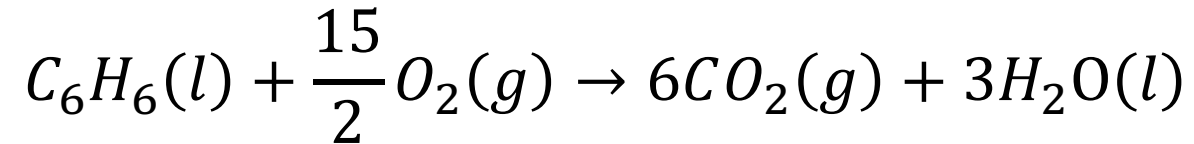
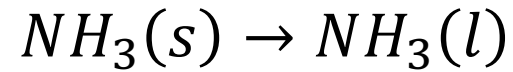
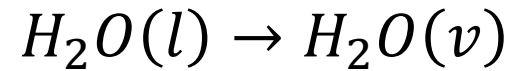




		<i>mean bond enthalpy (in kJ/mol)</i>
OH---O	H-bond	20
H-O	covalent bond	463
OH---N	H-bond	29
NH---N	H-bond	14
H-N	covalent bond	388



Sign of ΔH



Determination of Heat of Reaction



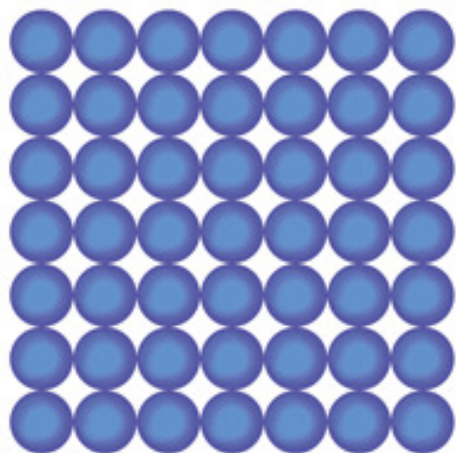
$$\Delta H = \Sigma \text{BE}(\text{reactants}) - \Sigma \text{BE}(\text{products})$$

Type of bonds broken	Number of bonds broken	Bond energy (kJ/mol)	Energy change (kJ)
C=C(g)	1	BE _{C=C}	BE _{C=C}
H — H	1	BE _{H-H}	BE _{H-H}
Type of bonds formed	Number of bonds formed	Bond energy (kJ/mol)	Energy change (kJ)
C — C	1	BE _{C-C}	BE _{C-C}
C — H	6-4=2	2BE _{C-H}	2BE _{C-H}

$$\Delta H = \text{BE}_{\text{C}=\text{C}} + \text{BE}_{\text{H}-\text{H}} - \text{BE}_{\text{C}-\text{C}} - 6\text{BE}_{\text{C}-\text{H}}$$

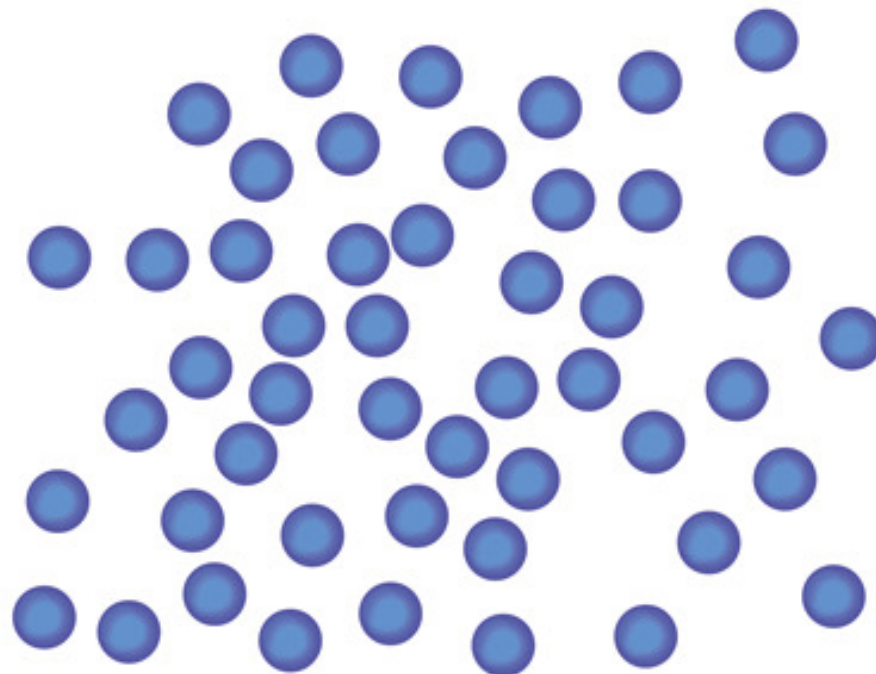
Entropy

Low Entropy

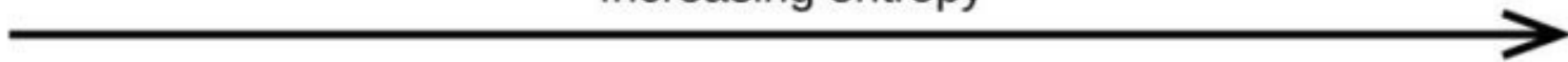


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Increase in Entropy

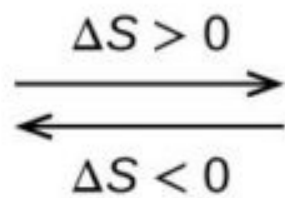
High Entropy



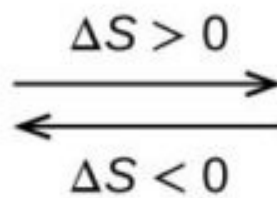
Increasing entropy



Crystalline solid

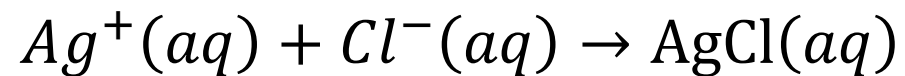
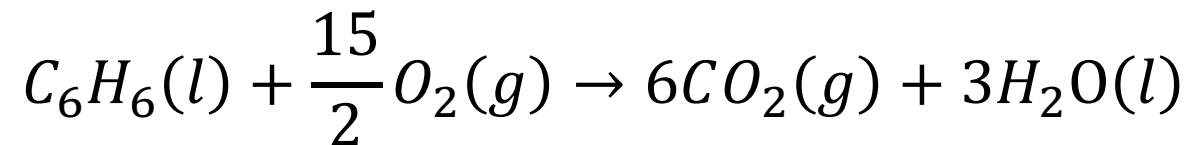
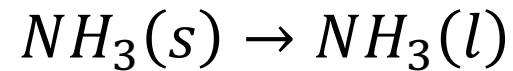
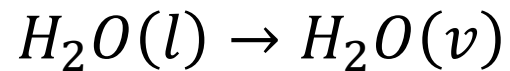


Liquid

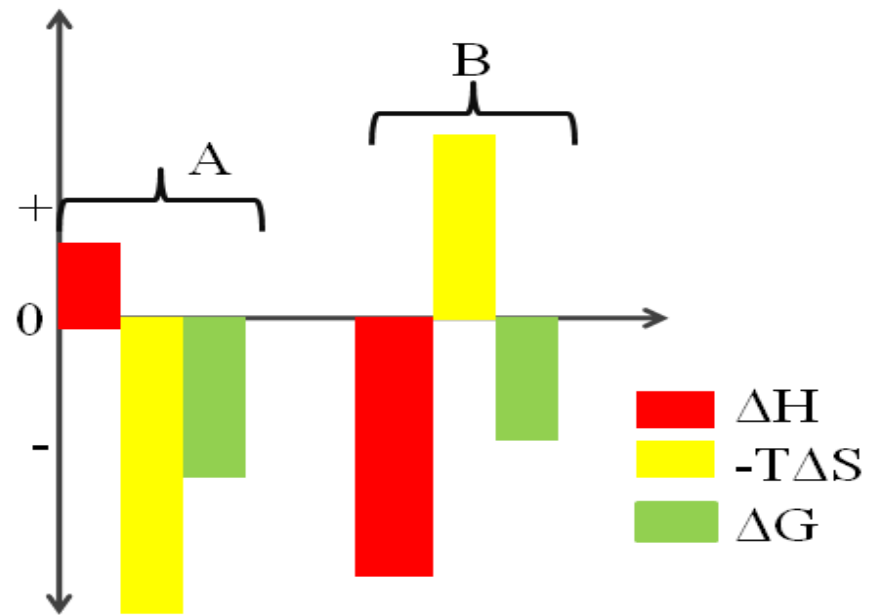


Gas

Sign of ΔS



Enthalpy/ Entropy driven process



A-Entropy driven
B-Enthalpy driven

Solid, liquid and Gas

Solid $\rightarrow$ liquid $\rightarrow$ Gas

The process of melting and vaporization both requires heat. Thus, these processes are endothermic reactions. $\Delta H > 0$

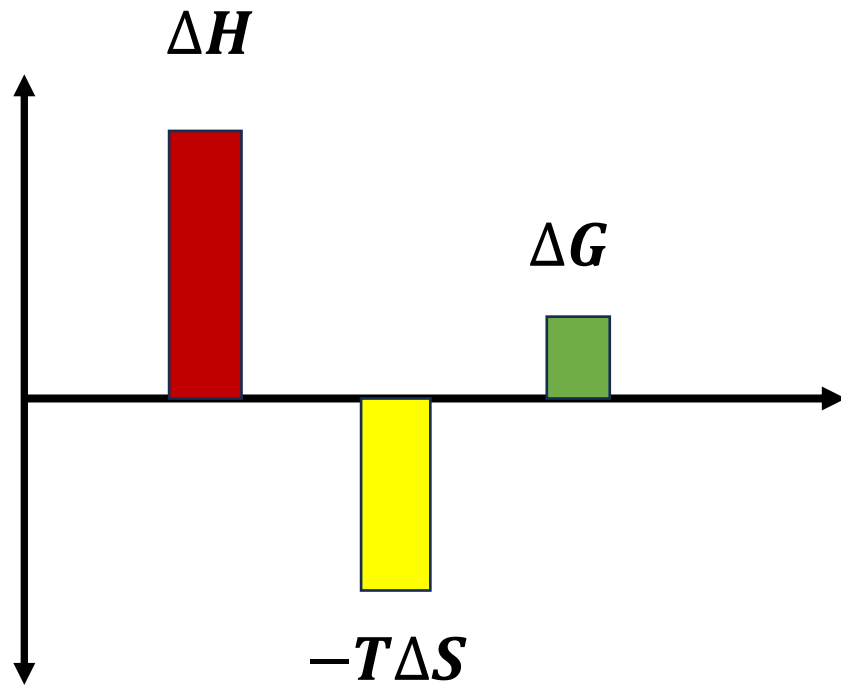
The molecules are more disordered in liquid than solid.

The molecules are more disordered in gas than a liquid.

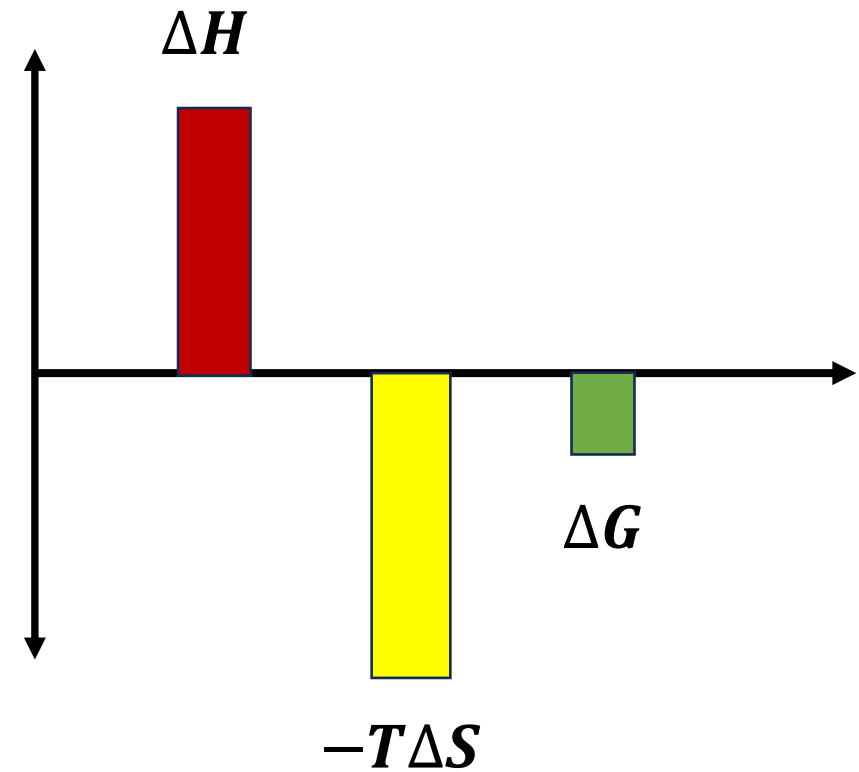
Therefore, entropy increases on melting and vaporization

$$\Delta H > 0 \text{ and } \Delta S > 0$$

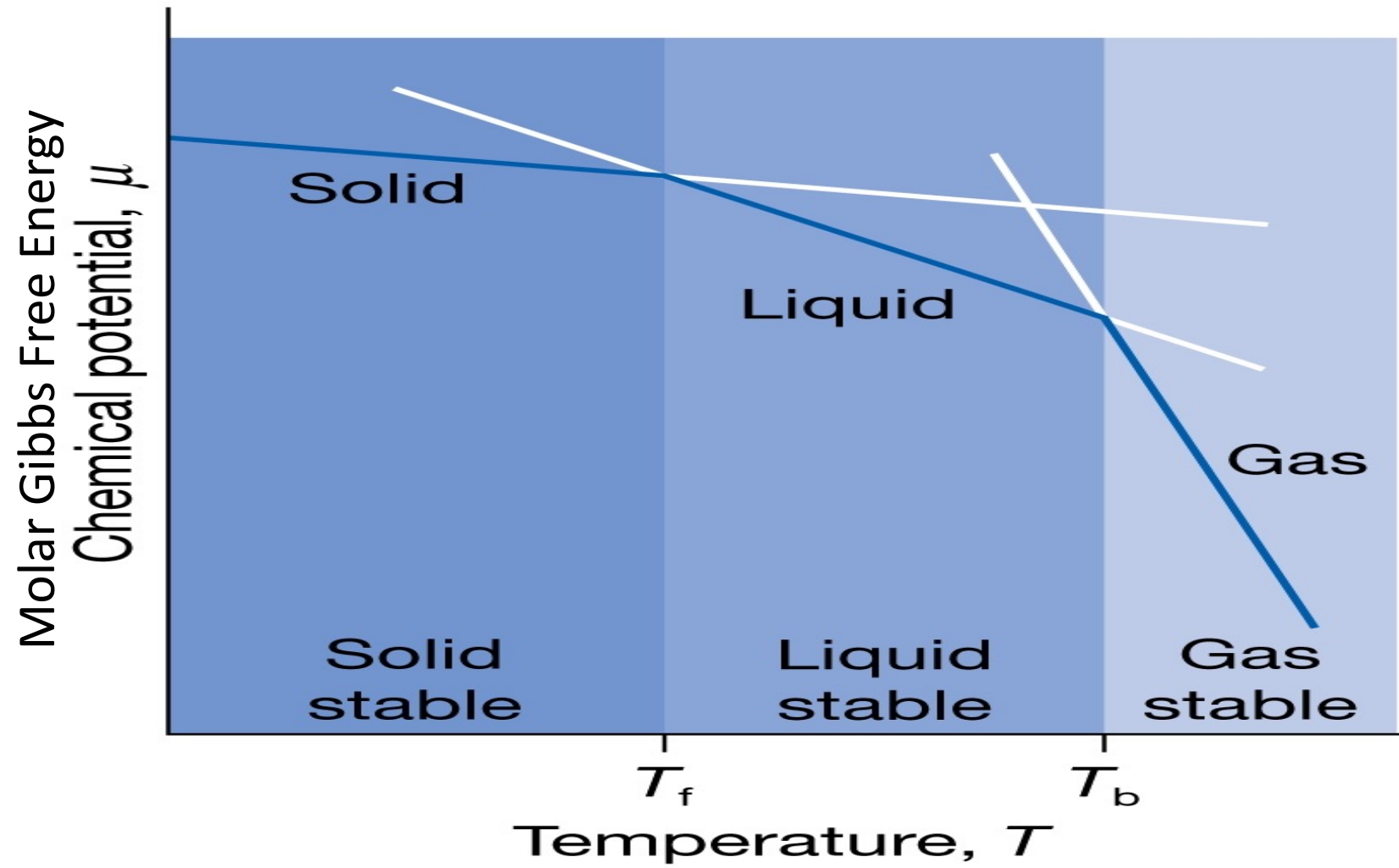
Solid to liquid to gas



Low Temperature



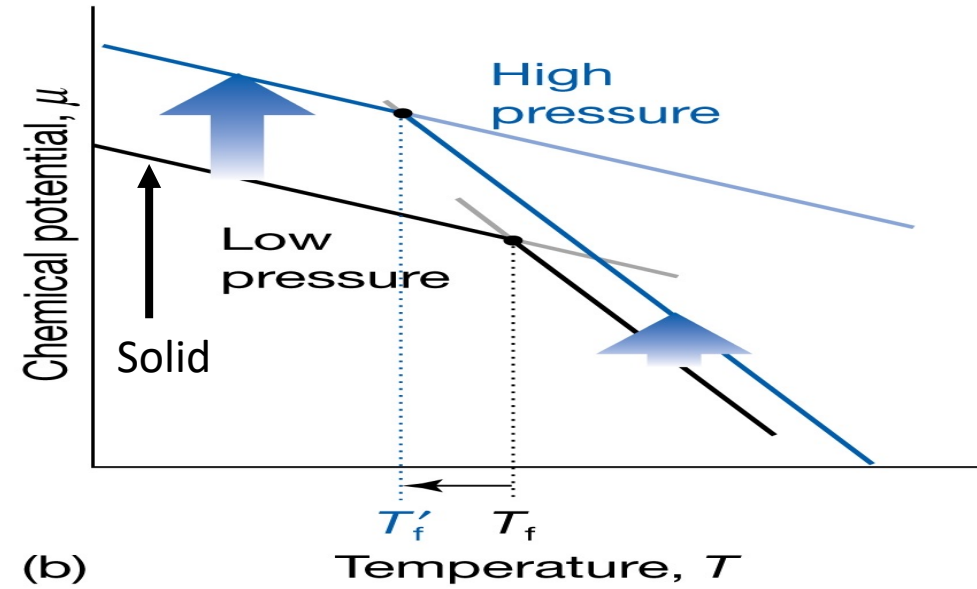
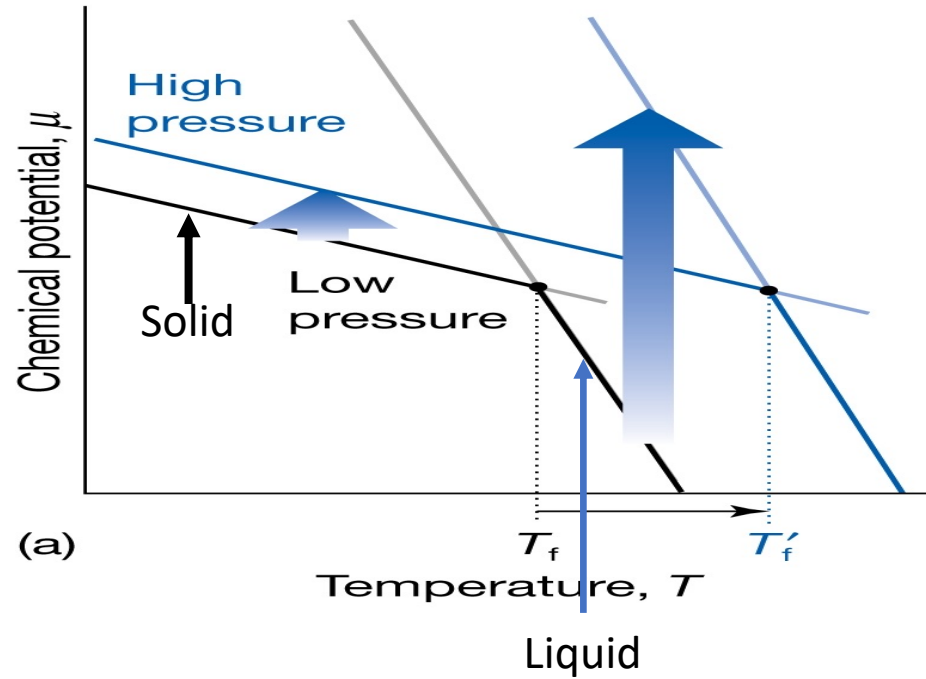
High Temperature



Effect of pressure of melting
point, freezing point

Response of melting temperature to the applied pressure

Water



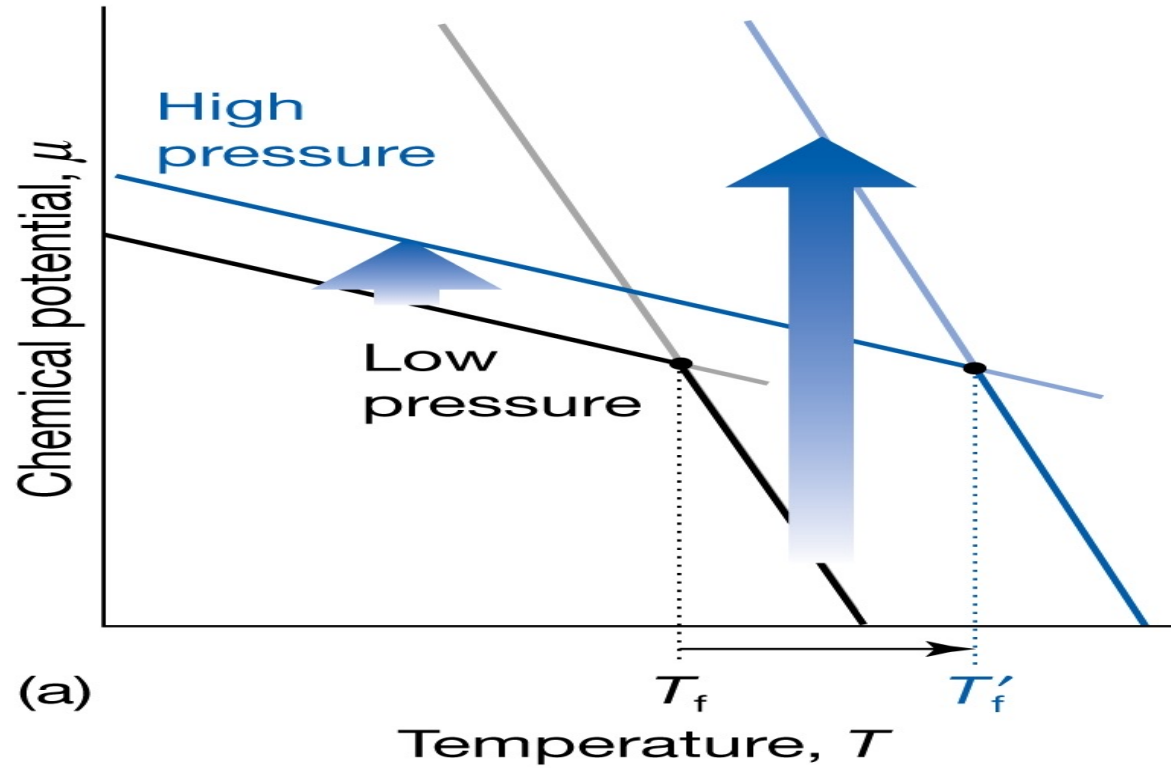
Ice-skating

- As we glide across the ice, we exert pressure on the thin blade, and are therefore creating a small stream of water in our path by melting that ice. The water between the blade and the ice is what we really glide across.



Effect of Pressure on Boiling point

Elevation of boiling point



Elevation of boiling point

- An application of this effect is the pressure cooker. A pressure cooker traps steam inside it, raising the pressure to about twice the atmospheric pressure. As a result, the water boils at $120\text{ }^{\circ}\text{C}$. Food cooks quicker at the higher temperature.



Autoclave

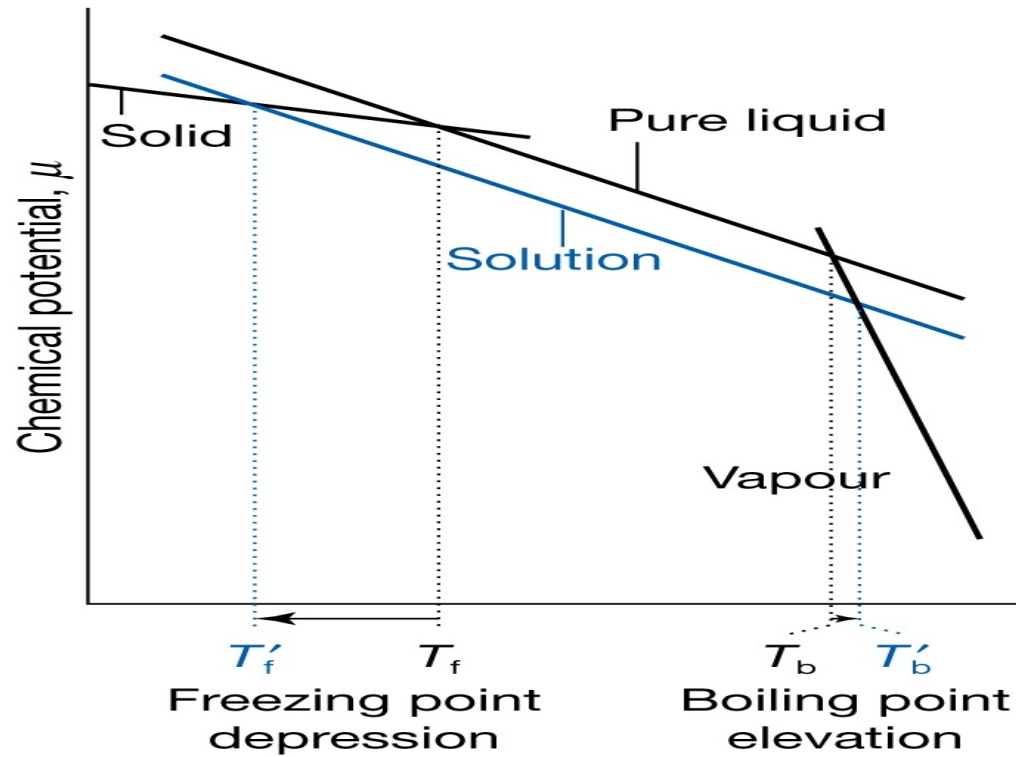
Sterilization of both materials and contaminated media is an essential process in a laboratory for in vitro cultures.

This sterilization is usually done in a device called an autoclave.

Essentially, an autoclave is a container that exposes the material to be sterilized to temperatures above the boiling point of water, which is achieved by increasing pressure.



Addition of a non-volatile
solute



There is elevation of boiling point upon addition of a solute.

There is depression of freezing point upon addition of a solute.